

Physical Chemistry (II)

Lecture 05

2D and 3D Particles in a Box



In the Last Two Classes

1. Wavefunction and its interpretation

2. Quantum measurement

- Observables and operators
- Measurement and collapse of wave function
- Uncertainty principle

3. Time evolution of wavefunction: Schrödinger equation

❖ Example: 1D particle in a box



Last class

Generalized Uncertainty Principle

$$\sigma_A^2 \sigma_B^2 \geq -\frac{1}{4} \langle [\hat{A}, \hat{B}] \rangle^2$$

e.g. position-momentum uncertainty

$$[\hat{x}, \hat{p}_x] = \hat{x}\hat{p}_x - \hat{p}_x\hat{x} = x \frac{\hbar}{i} \frac{\partial}{\partial x} - \frac{\hbar}{i} \frac{\partial}{\partial x} x = i\hbar$$

$$\sigma_x^2 \sigma_{p_x}^2 \geq -\frac{1}{4} \langle i\hbar \rangle^2 = \left(\frac{\hbar}{2} \right)^2$$

Last class

Time Evolution of Wavefunction

$$i\hbar \frac{\partial}{\partial t} \Psi(x, y, z, t) = \hat{H} \Psi(x, y, z, t)$$

where

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + \mathcal{V}(\hat{x}, \hat{y}, \hat{z}, t)$$

(Time-dependent Schrödinger equation)

Time-Independent Potential

- Time-independent Schrödinger equation

$$\hat{H}\psi(x, y, z) = E\psi(x, y, z)$$

→ Eigenvalue equation for the Hamiltonian

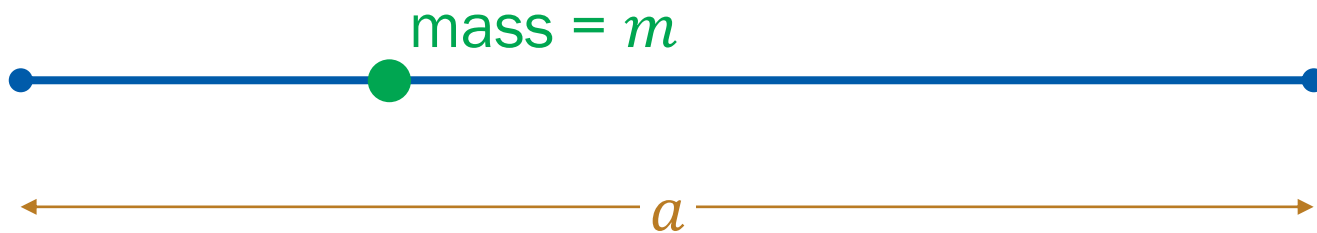
- Temporal solution: $T(t) = e^{-iEt/\hbar}$

$$|T(t)|^2 = 1$$

$$|\Psi(x, y, z, t)|^2 = |\psi(x, y, z)|^2$$

Last class

Particle in a 1D Box



$$V(x) = \begin{cases} \infty, & x < 0 \\ 0, & 0 \leq x \leq a \\ \infty, & x > a \end{cases}$$

Final Solutions

- Index $n = 1, 2, 3, \dots$ (“quantum number”)
- Time-independent wavefunction

$$\psi_n(x) = \sqrt{2/a} \sin\left(\frac{n\pi x}{a}\right), \quad 0 \leq x \leq a$$

- Energy

$$E_n = \frac{h^2}{8ma^2} n^2$$

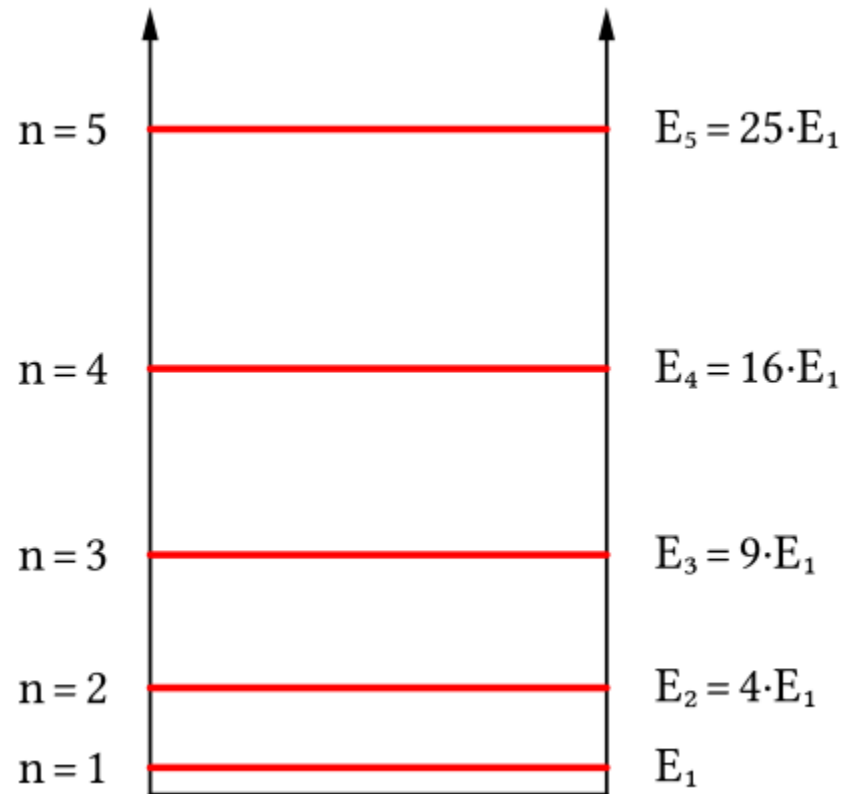
- Total wavefunction

$$\Psi_n(x, t) = \psi_n(x)T_n(t) = \sqrt{2/a} \sin\left(\frac{n\pi x}{a}\right) e^{-iE_n t/\hbar}$$

Last class

Energy Levels

$$E_n = \frac{h^2}{8ma^2} n^2$$



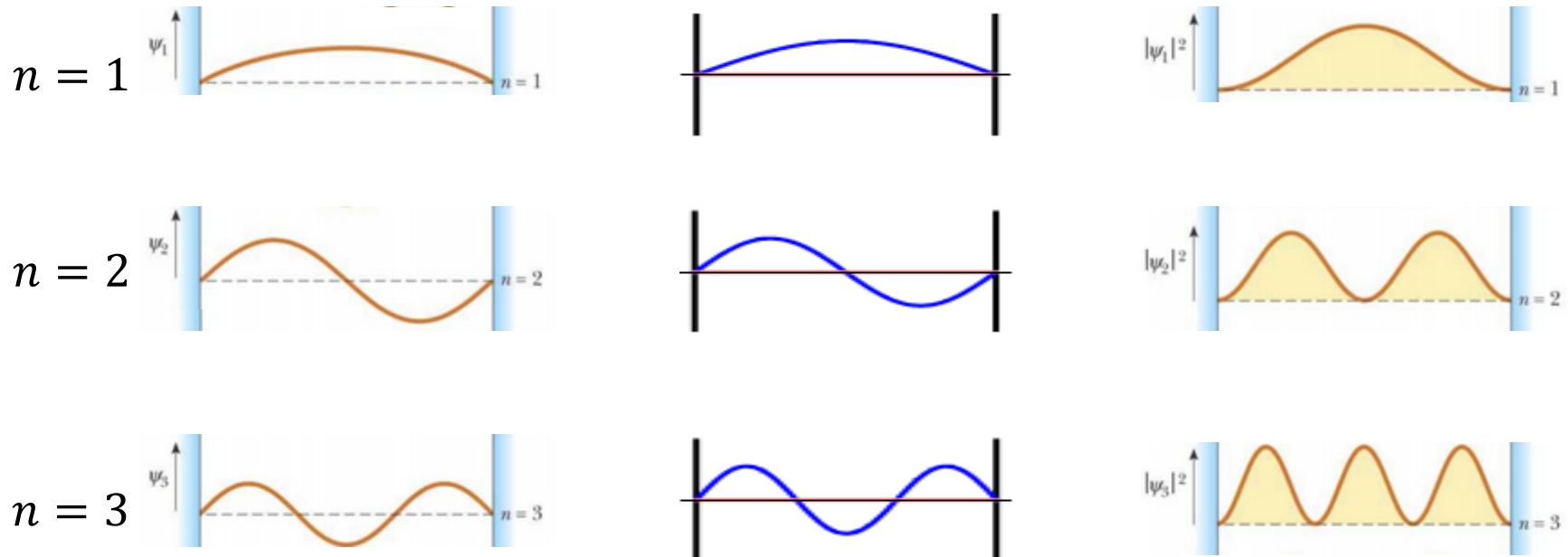
Last class

Wavefunctions

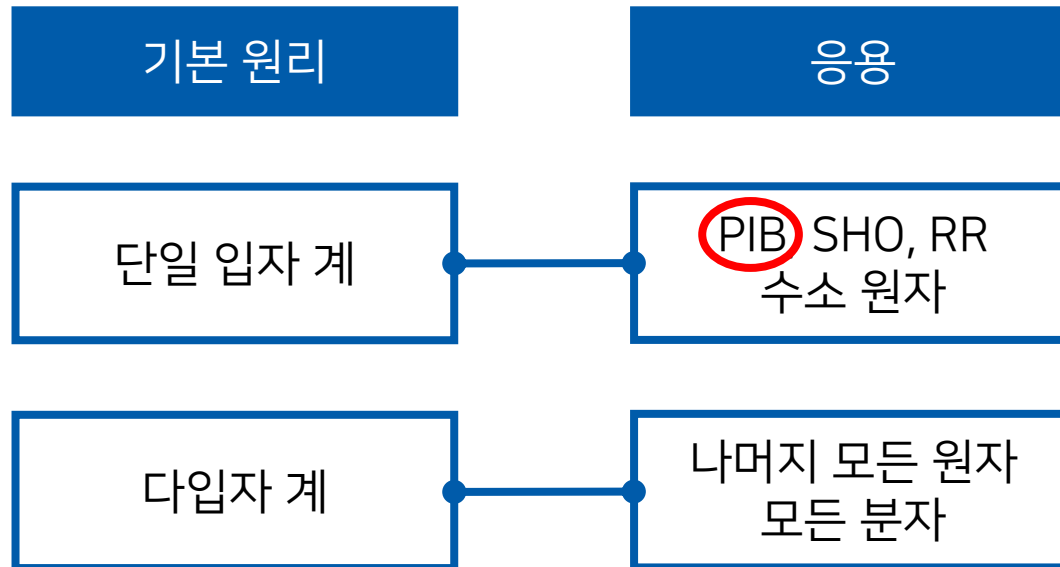
$$\psi_n(x)$$

$$\Psi_n(x, t)$$

$$|\Psi_n(x, t)|^2$$

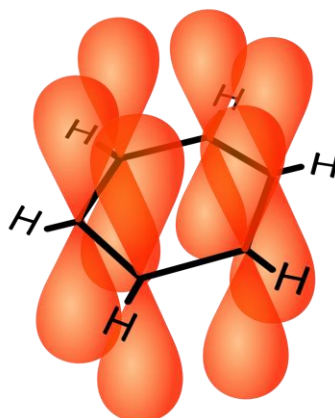
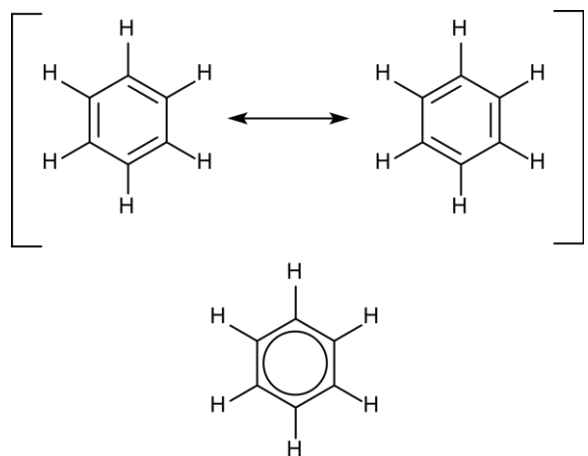


Where Are We?

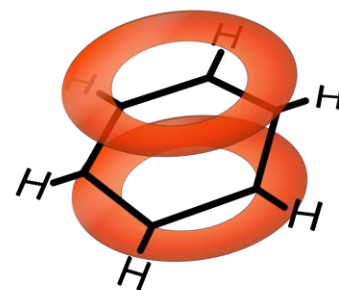


RS 3.4
M 3.5

Application: Conjugated System

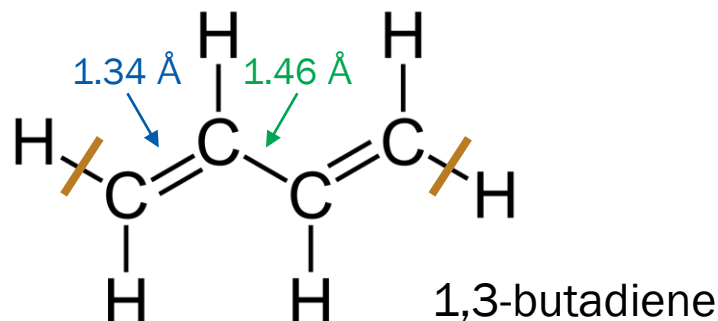


6 p-orbitals



delocalized

Free Electron Model



Electron considered as a particle confined in a box of size

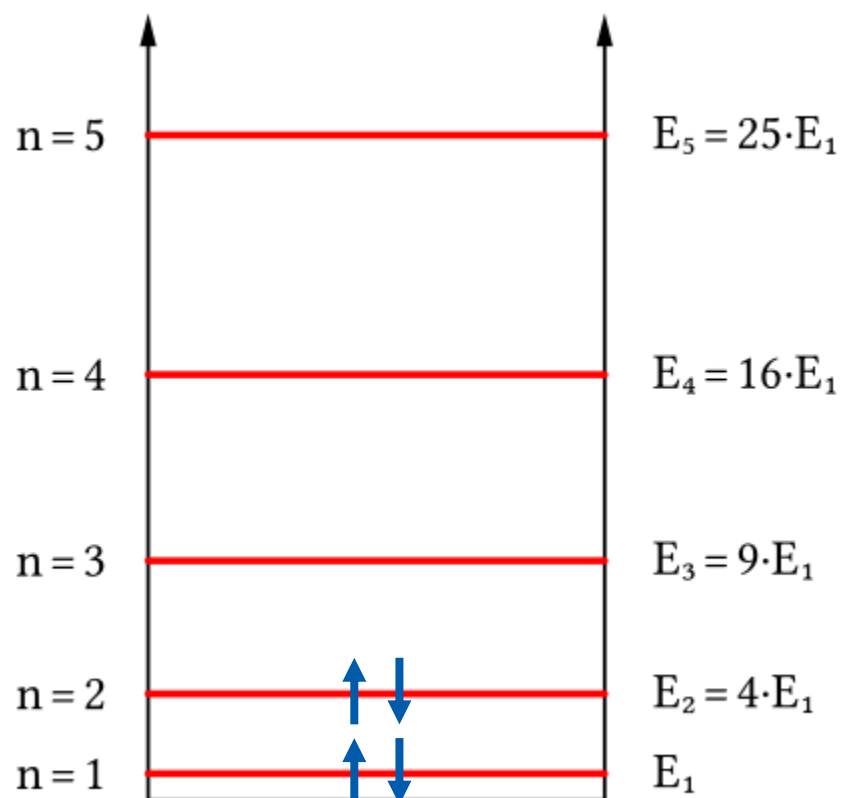
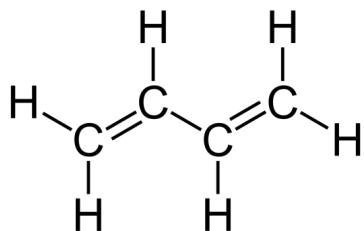
$$a = (1.46 \times 0.5 + 1.34 + 1.46 + 1.34 + 1.46 \times 0.5) \text{ \AA} = 5.6 \text{ \AA}$$

The particle in a box model gives

$$E_n = \frac{h^2}{8ma^2} n^2 = (1.92 \times 10^{-19} \text{ J}) n^2$$

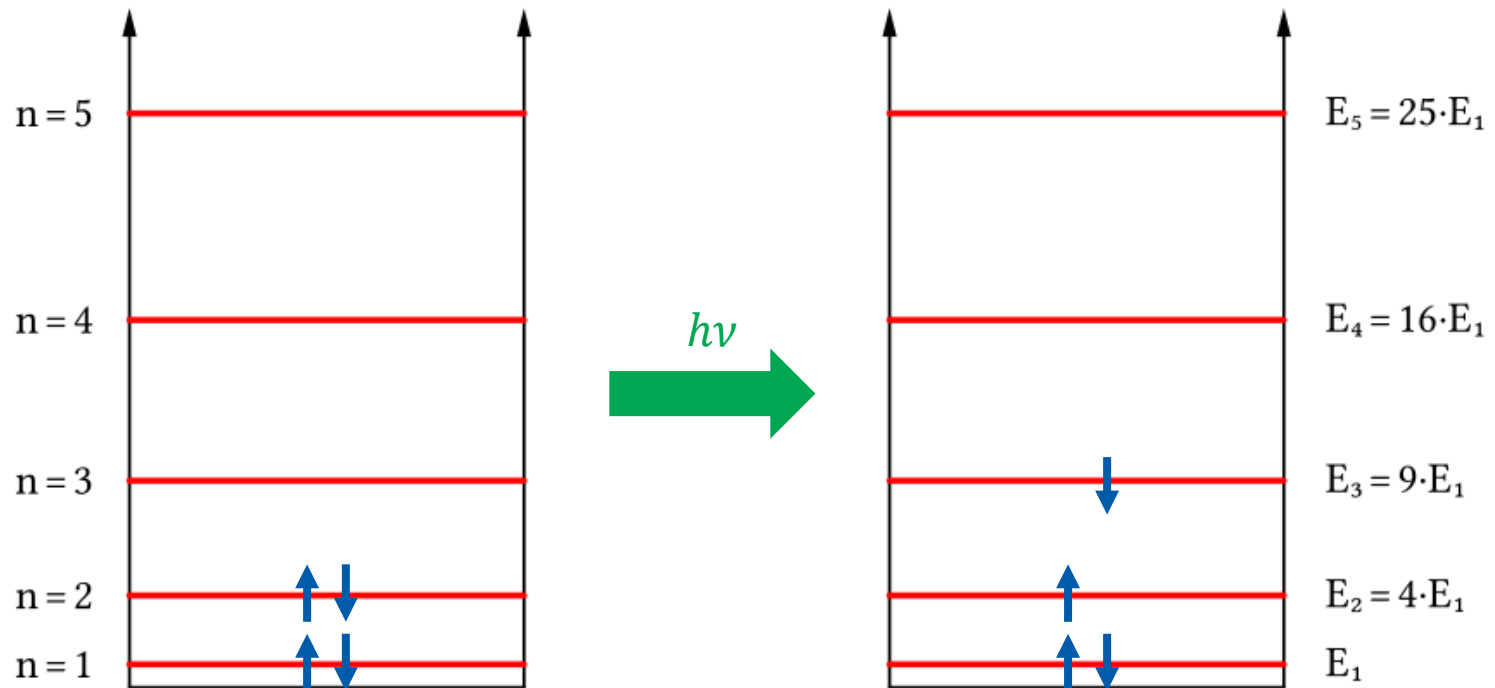
RS 3.4
M 3.5

Energy Levels of 1,3-Butadiene



RS 3.4
M 3.5

Absorption Spectrum



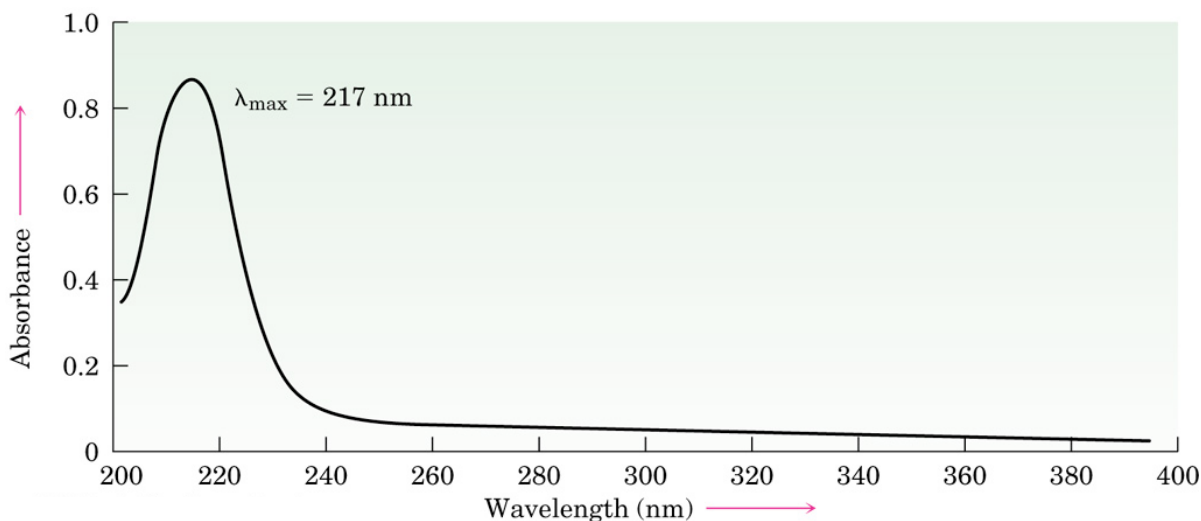
$$h\nu = \Delta E = E_3 - E_2 = 5E_1 = 9.60 \times 10^{-19} \text{ J}$$

Prediction vs. Experiment

$$h\nu = 9.60 \times 10^{-19} \text{ J}$$

$$\nu = 1.45 \times 10^{15} \text{ Hz}$$

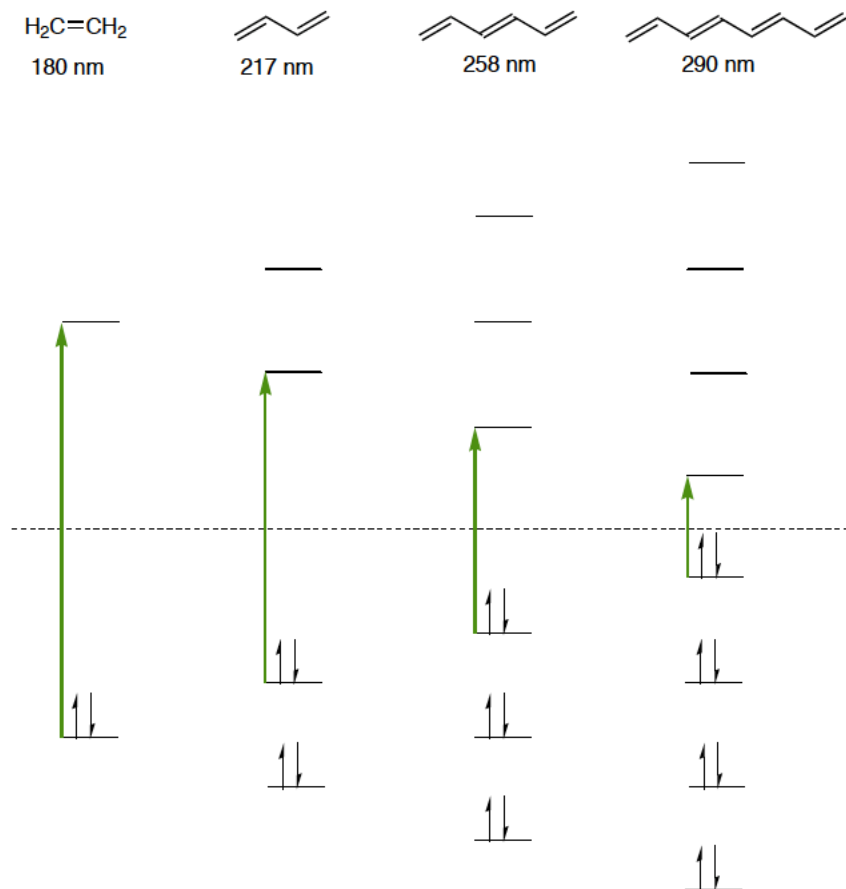
$$\lambda = \frac{c}{\nu} = 2.06 \times 10^{-7} \text{ m} = 206 \text{ nm}$$



(Image: <https://www.vanderbilt.edu/AnS/Chemistry/Rizzo/chem220a/Ch14slides.pdf>)

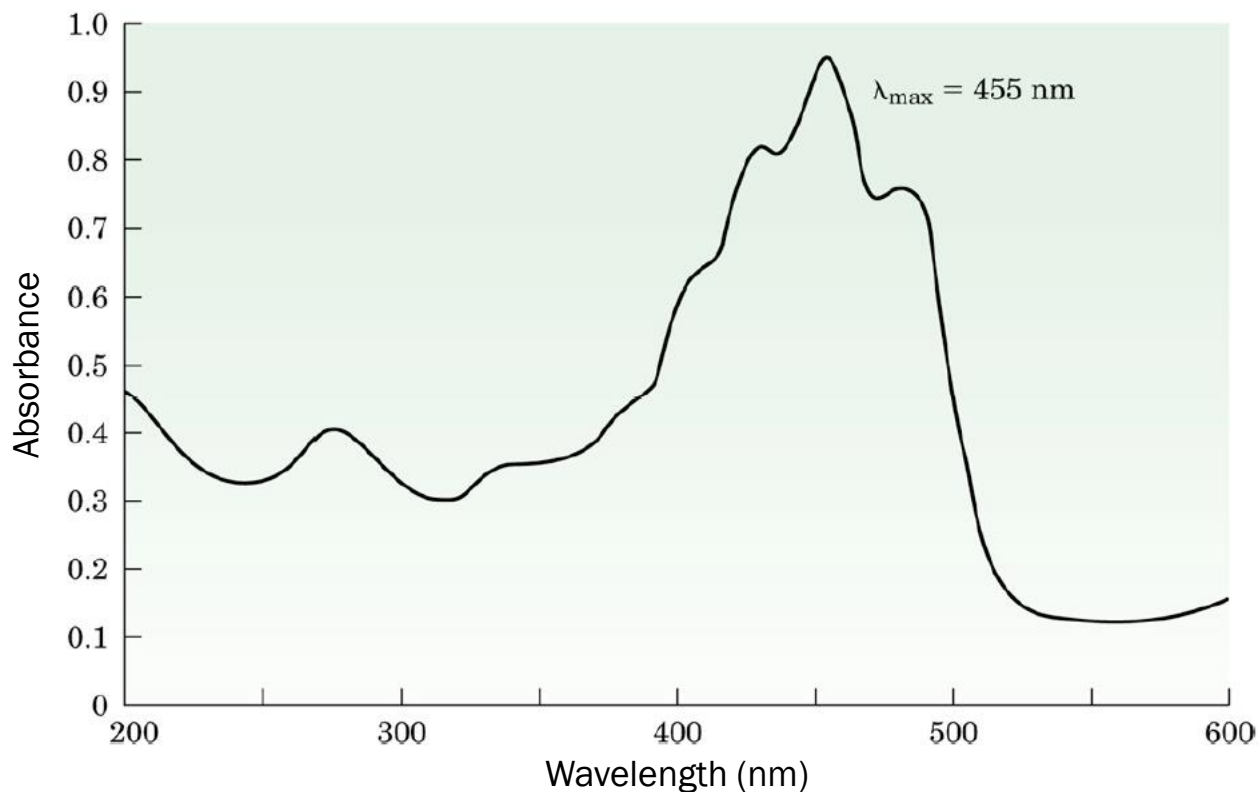
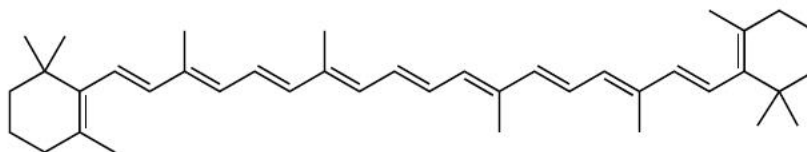
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Size Effect



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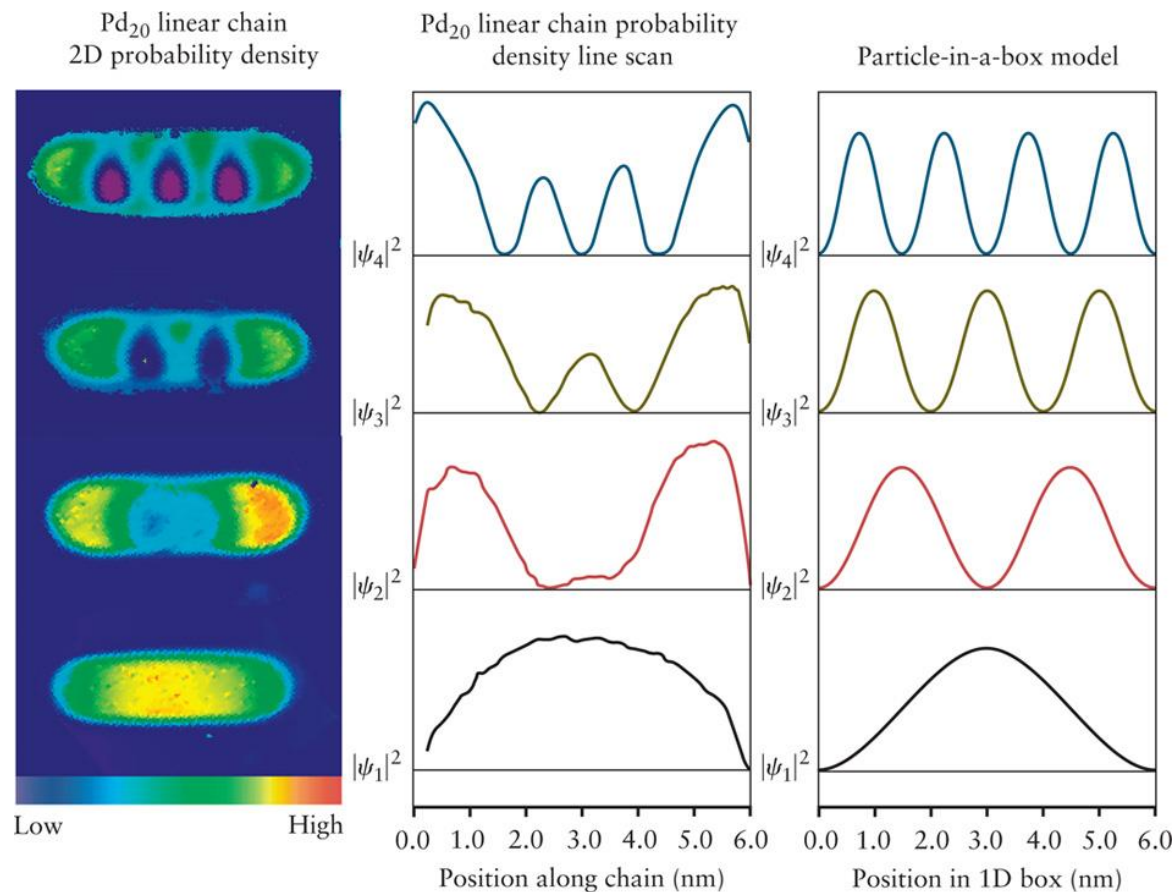
Beta-Carotene



(Image: <https://www.vanderbilt.edu/AnS/Chemistry/Rizzo/chem220a/Ch14slides.pdf>)

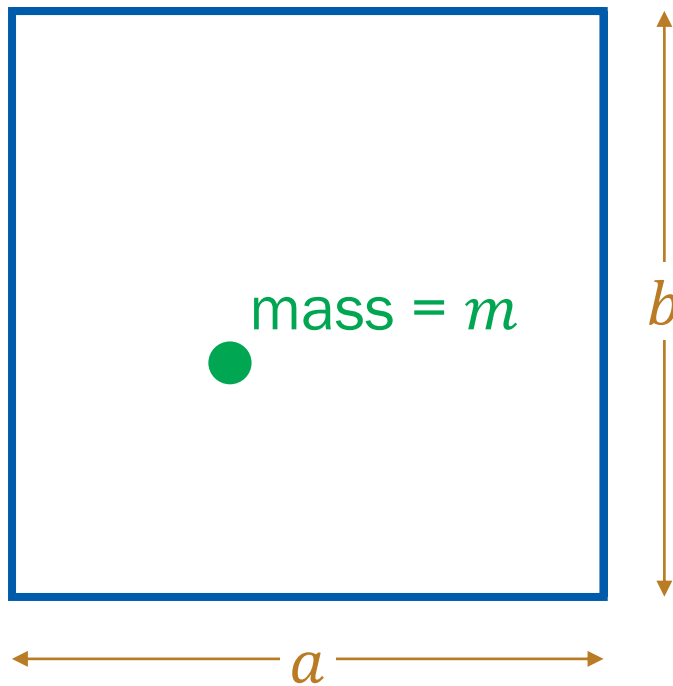
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Another Application: Pd Chain



(Image: Oxtoby et al., *Principles of Modern Chemistry*, 7th Ed. (2011), Fig. 4.27)

2D Particle in a Box



$$\mathcal{V}(x, y) = \mathcal{V}_x(x)\mathcal{V}_y(y)$$

$$\mathcal{V}_x(x) = \begin{cases} \infty, & x < 0 \\ 0, & 0 \leq x \leq a \\ \infty, & x > a \end{cases}$$
$$\mathcal{V}_y(y) = \begin{cases} \infty, & y < 0 \\ 0, & 0 \leq y \leq b \\ \infty, & y > b \end{cases}$$

Time-Indep. Schrödinger Equation

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + \mathcal{V}(x, y) \right] \psi(x, y) = E\psi(x, y)$$

For $0 \leq x \leq a$ and $0 \leq y \leq b$,

$$-\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \psi(x, y) = E\psi(x, y)$$

Let $\psi(x, y) = X(x)Y(y)$; rearrangement leads to

$$-\frac{\hbar^2}{2m} \left(\frac{1}{X} \frac{\partial^2 X}{\partial x^2} + \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} \right) = E$$

Separation of Variables

$$-\frac{\hbar^2}{2m} \left(\frac{1}{X} \frac{\partial^2 X}{\partial x^2} + \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} \right) = E$$

$$\underbrace{-\frac{\hbar^2}{2m} \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2}}_{\text{function of } y} = \underbrace{E + \frac{\hbar^2}{2m} \frac{1}{X} \frac{\partial^2 X}{\partial x^2}}_{\text{function of } x} = E_y$$

System of Equations

$$-\frac{\hbar^2}{2m} \frac{1}{X} \frac{\partial^2 X}{\partial x^2} = E - E_y = E_x$$

$$-\frac{\hbar^2}{2m} \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} = E_y$$

Each of these equations is identical with
the equation for a 1D particle in a box

Solutions

- Quantum numbers $n_x, n_y = 1, 2, 3, \dots$
- Energy levels

$$E_x = \frac{h^2}{8ma^2} n_x^2, \quad E_y = \frac{h^2}{8mb^2} n_y^2$$

$$E = E_x + E_y = \frac{h^2}{8m} \left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} \right) = E_{(n_x, n_y)}$$

Solutions

- (Time-independent) Wavefunctions

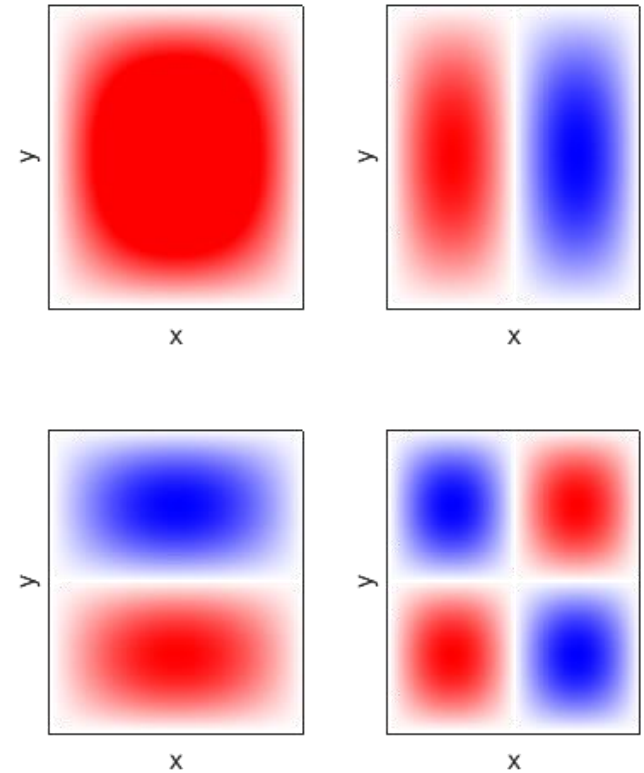
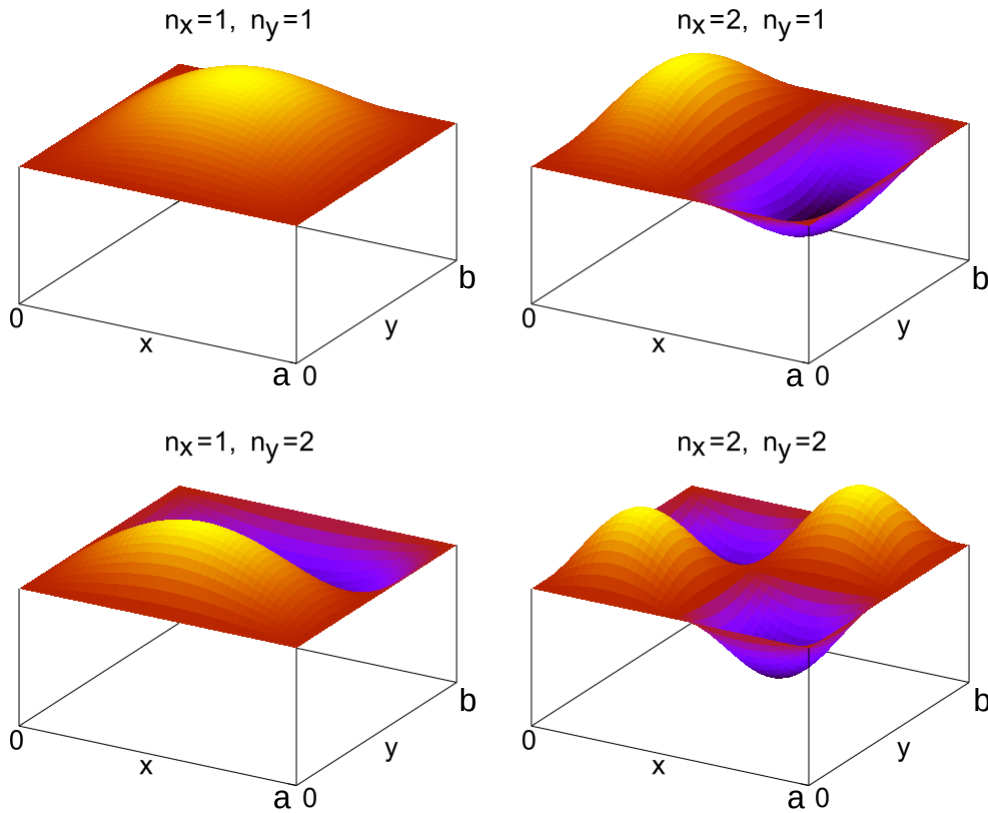
$$X(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n_x \pi x}{a}\right), \quad 0 \leq x \leq a$$

$$Y(y) = \sqrt{\frac{2}{b}} \sin\left(\frac{n_y \pi y}{b}\right), \quad 0 \leq y \leq b$$

$$\psi_{n_x, n_y}(x, y) = \sqrt{\frac{4}{ab}} \sin\left(\frac{n_x \pi x}{a}\right) \sin\left(\frac{n_y \pi y}{b}\right)$$

A (area)

Wavefunctions

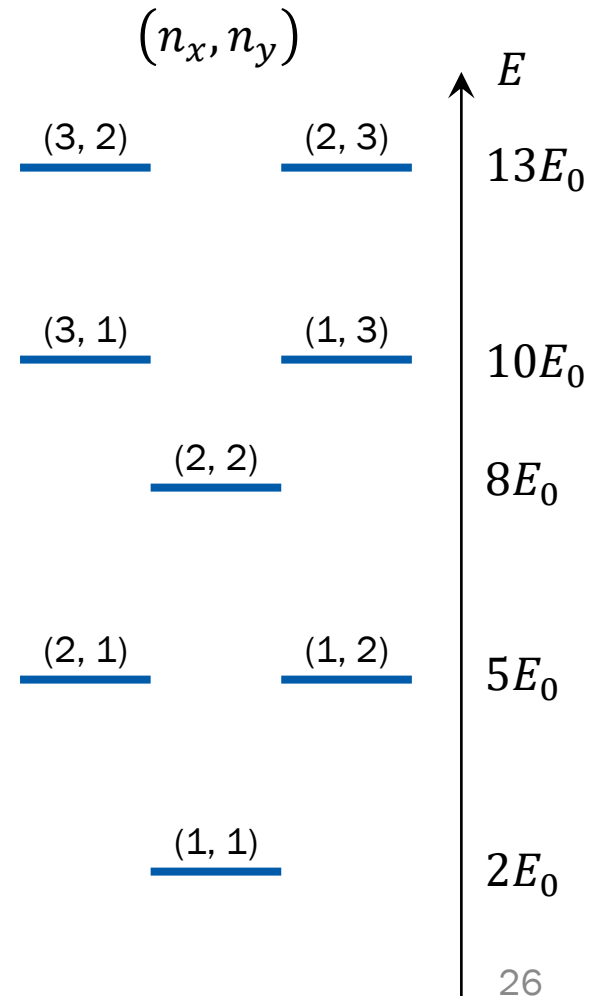


Degeneracy

When $a = b$,

$$E = \frac{h^2}{8ma^2} (n_x^2 + n_y^2) = E_0 (n_x^2 + n_y^2)$$

- **Degeneracy:** different states have the same energy
- The states are called **degenerate**



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When You Have Degeneracy

The **linear combination** of wavefunctions of degenerate states **also satisfies the Schrödinger equation**

For ψ_1 and ψ_2 such that $\hat{H}\psi_1 = E_0\psi_1$ and $\hat{H}\psi_2 = E_0\psi_2$, we can define $\phi = c_1\psi_1 + c_2\psi_2$ and obtain

$$\hat{H}\phi = \hat{H}(c_1\psi_1 + c_2\psi_2) = c_1E_0\psi_1 + c_2E_0\psi_2 = E_0\phi$$

Example

The states (2, 1) and (1, 2) are degenerate (energy = $5E_0$)

$$\psi_{2,1}(x, y) = \frac{2}{a} \sin\left(\frac{2\pi x}{a}\right) \sin\left(\frac{\pi y}{a}\right)$$

$$\psi_{1,2}(x, y) = \frac{2}{a} \sin\left(\frac{\pi x}{a}\right) \sin\left(\frac{2\pi y}{a}\right)$$

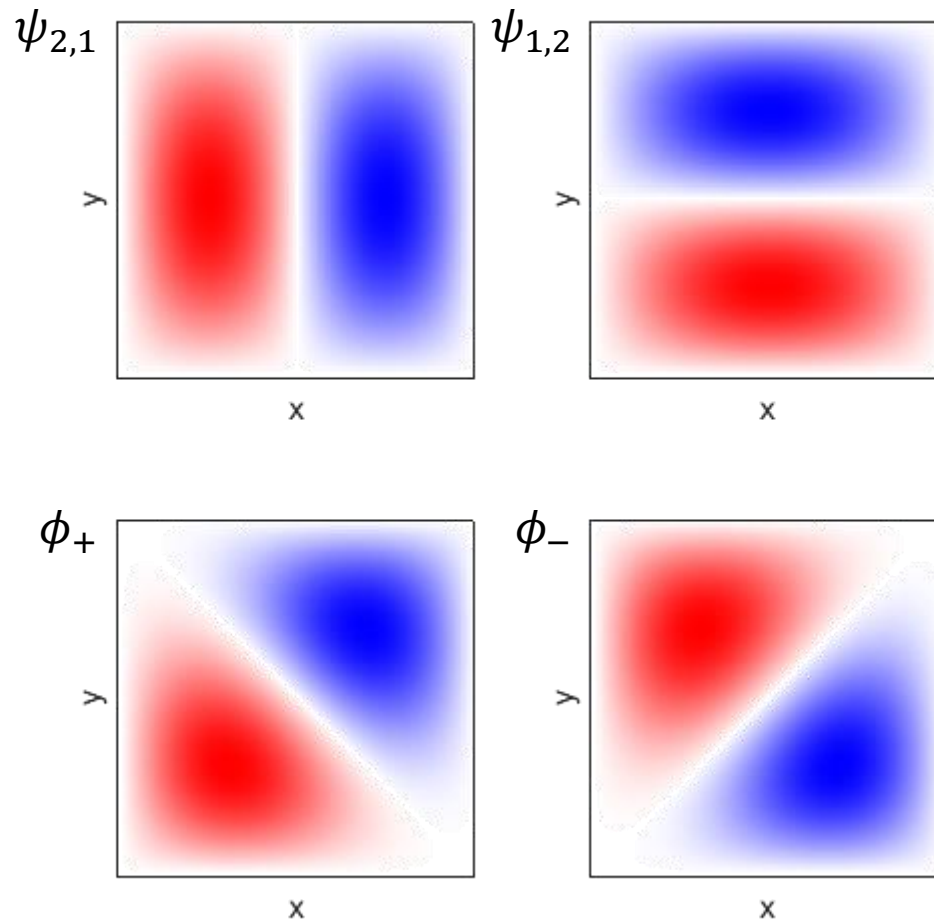
We can make

$$\phi_+ = \frac{1}{\sqrt{2}}(\psi_{2,1} + \psi_{1,2}); \quad \phi_- = \frac{1}{\sqrt{2}}(\psi_{2,1} - \psi_{1,2})$$

 which are also the solutions of the Schrödinger equation

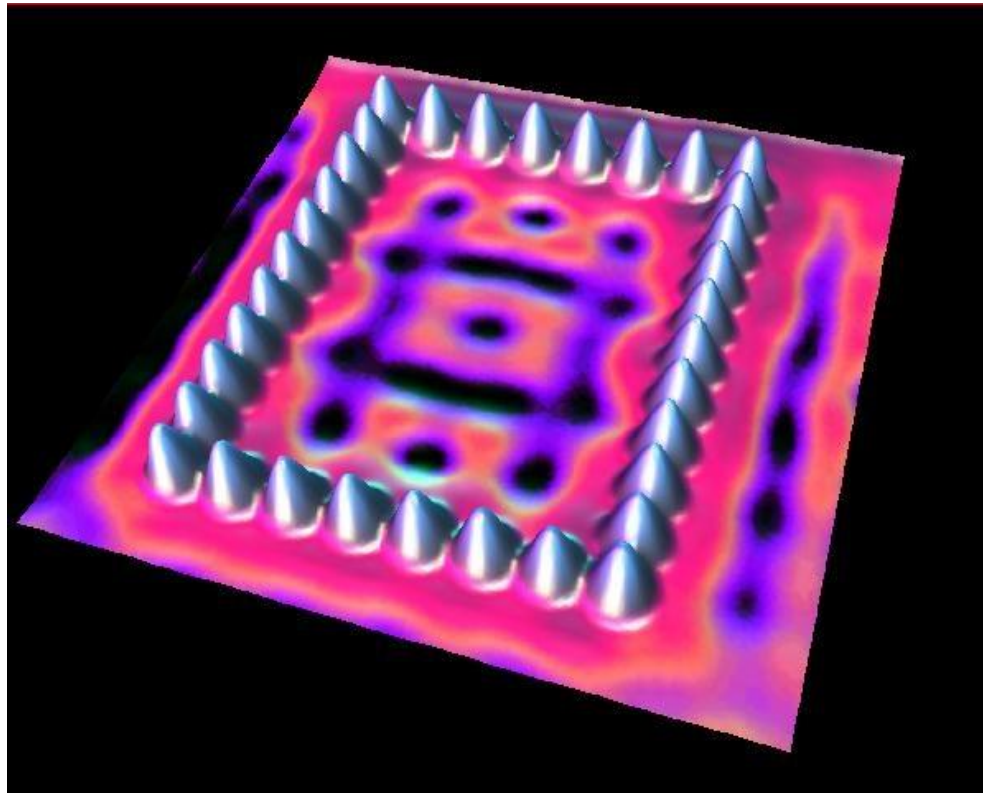
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Example



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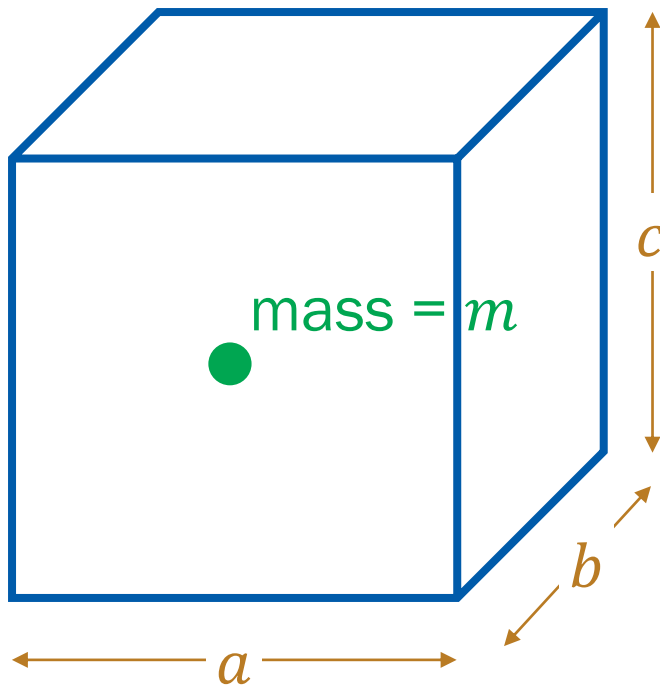
Application: “Quantum Square”



STM image for Fe atoms on a Cu(111) surface

(Image: <https://www.pinterest.co.kr/pin/469641067361529171/>)

3D Particle in a Box



$$\mathcal{V}(x, y, z) = \mathcal{V}_x(x)\mathcal{V}_y(y)\mathcal{V}_z(z)$$

$$\mathcal{V}_x(x) = \begin{cases} \infty, & x < 0 \\ 0, & 0 \leq x \leq a \\ \infty, & x > a \end{cases}$$

$$\mathcal{V}_y(y) = \begin{cases} \infty, & y < 0 \\ 0, & 0 \leq y \leq b \\ \infty, & y > b \end{cases}$$

$$\mathcal{V}_z(z) = \begin{cases} \infty, & z < 0 \\ 0, & 0 \leq z \leq c \\ \infty, & z > c \end{cases}$$

Time-Indep. Schrödinger Equation

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + \mathcal{V}(x, y, z) \right] \psi(x, y, z) = E\psi(x, y, z)$$

For $0 \leq x \leq a, 0 \leq y \leq b, 0 \leq z \leq c,$

$$-\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \psi(x, y, z) = E\psi(x, y, z)$$

Let $\psi(x, y, z) = X(x)Y(y)Z(z)$; rearrangement leads to

$$-\frac{\hbar^2}{2m} \left(\frac{1}{X} \frac{\partial^2 X}{\partial x^2} + \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} + \frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} \right) = E$$

Separation of Variables

$$-\frac{\hbar^2}{2m} \left(\frac{1}{X} \frac{\partial^2 X}{\partial x^2} + \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} + \frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} \right) = E$$

$$\underbrace{-\frac{\hbar^2}{2m} \frac{1}{Z} \frac{\partial^2 Z}{\partial z^2}}_{\text{function of } z} = E + \underbrace{\frac{\hbar^2}{2m} \left(\frac{1}{X} \frac{\partial^2 X}{\partial x^2} + \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} \right)}_{\text{function of } x, y} = E_z$$

Similarly, the right-handed side gives

$$-\frac{\hbar^2}{2m} \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} = (E - E_z) + \frac{\hbar^2}{2m} \frac{1}{X} \frac{\partial^2 X}{\partial x^2} = E_y$$

System of Equations

$$-\frac{\hbar^2}{2m} \frac{1}{X} \frac{\partial^2 X}{\partial x^2} = E - E_y - E_z = E_x$$

$$-\frac{\hbar^2}{2m} \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} = E_y$$

$$-\frac{\hbar^2}{2m} \frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} = E_z$$

Each of these equations is identical with the equation
for a 1-dimensional particle in a box

Solutions

- Quantum numbers $n_x, n_y, n_z = 1, 2, 3, \dots$
- Energy levels

$$E_x = \frac{h^2}{8ma^2} n_x^2, \quad E_y = \frac{h^2}{8mb^2} n_y^2, \quad E_z = \frac{h^2}{8mc^2} n_z^2$$

$$E = E_x + E_y + E_z = \frac{h^2}{8m} \left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} + \frac{n_z^2}{c^2} \right) = E_{(n_x, n_y, n_z)}$$

Solutions

- Wavefunctions

$$X(x) = \sqrt{2/a} \sin\left(\frac{n_x \pi x}{a}\right), \quad 0 \leq x \leq a$$

$$Y(y) = \sqrt{2/b} \sin\left(\frac{n_y \pi y}{b}\right), \quad 0 \leq y \leq b$$

$$Z(z) = \sqrt{2/c} \sin\left(\frac{n_z \pi z}{c}\right), \quad 0 \leq z \leq c$$

$$\psi(x, y, z) = \sqrt{\frac{8}{abc}} \sin\left(\frac{n_x \pi x}{a}\right) \sin\left(\frac{n_y \pi y}{b}\right) \sin\left(\frac{n_z \pi z}{c}\right)$$

V (volume)

Degeneracy

$$E_{(n_x, n_y, n_z)} = \frac{h^2}{8m} \left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} + \frac{n_z^2}{c^2} \right)$$

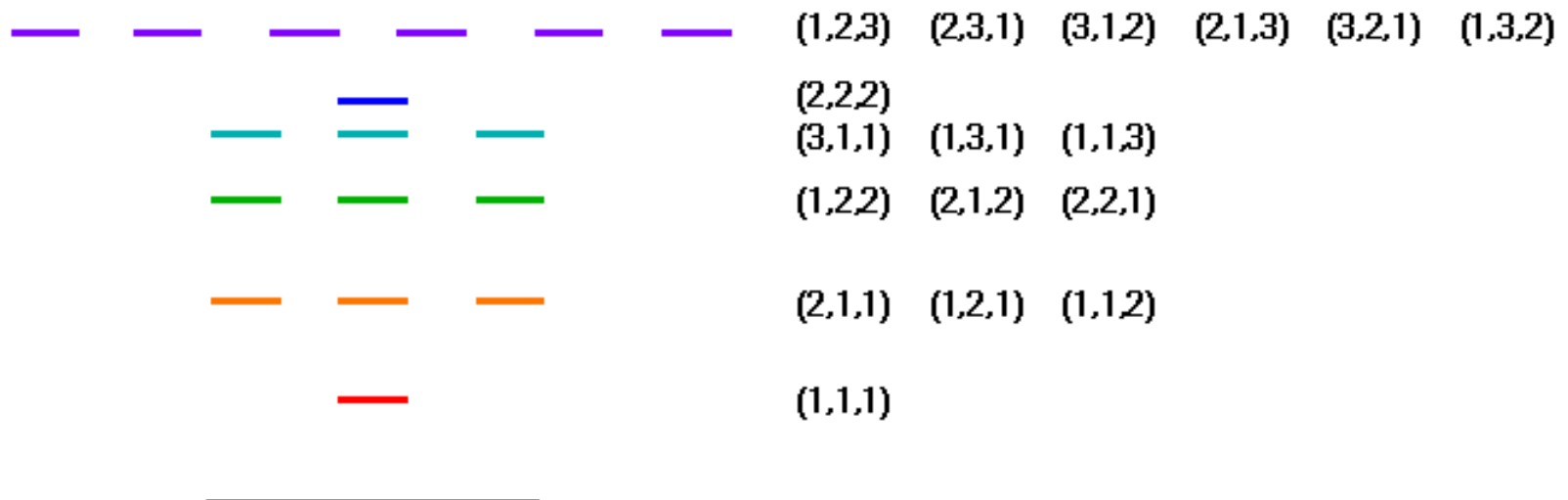
If $a = b = c$,

$$E_{(n_x, n_y, n_z)} = \frac{h^2}{8ma^2} (n_x^2 + n_y^2 + n_z^2)$$

Different states (= different combinations of n_x, n_y, n_z) can have the same energy → **degenerate states**

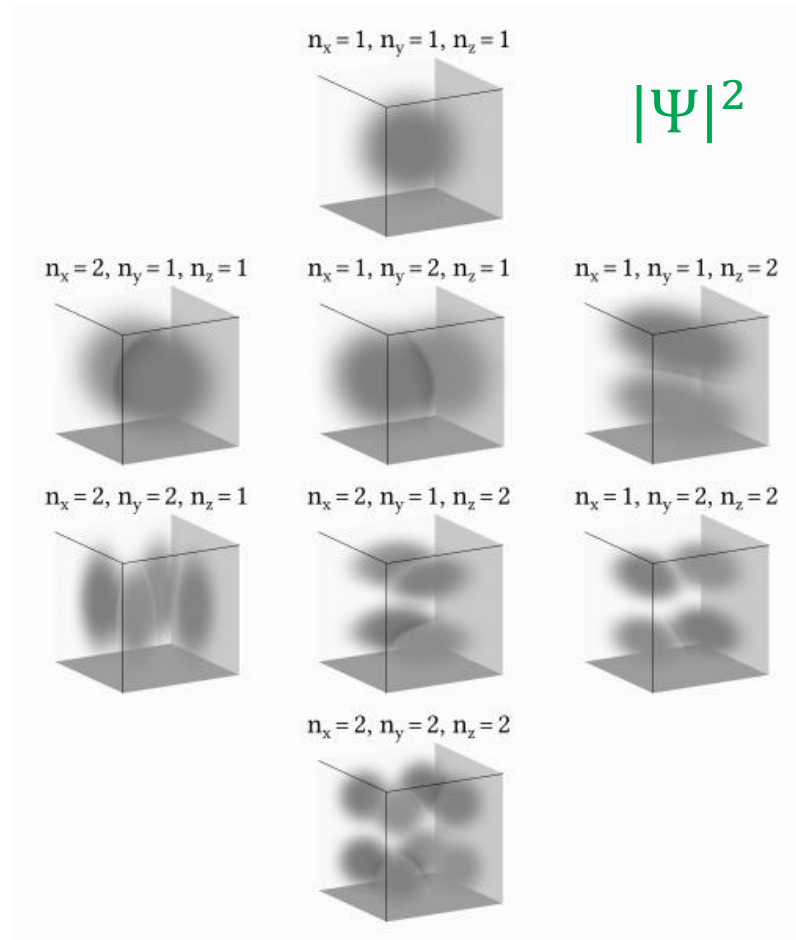
Energy Levels

$$E_{(n_x, n_y, n_z)} = \frac{h^2}{8ma^2} (n_x^2 + n_y^2 + n_z^2)$$



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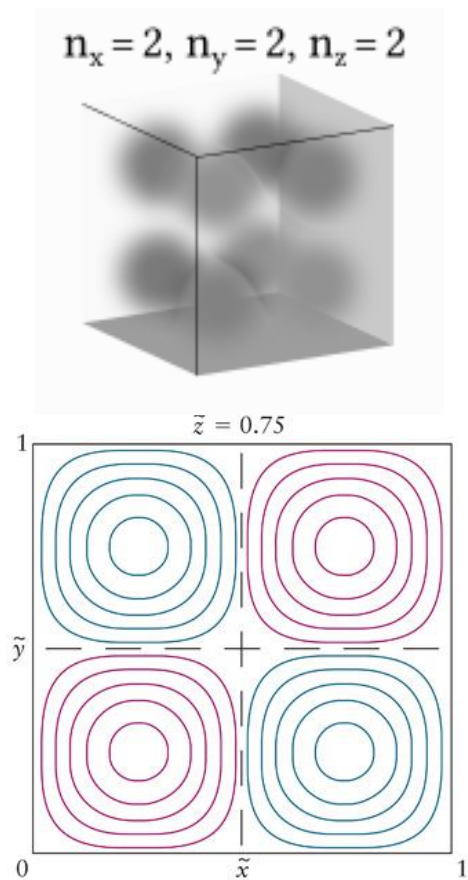
How to Visualize 3D WF?



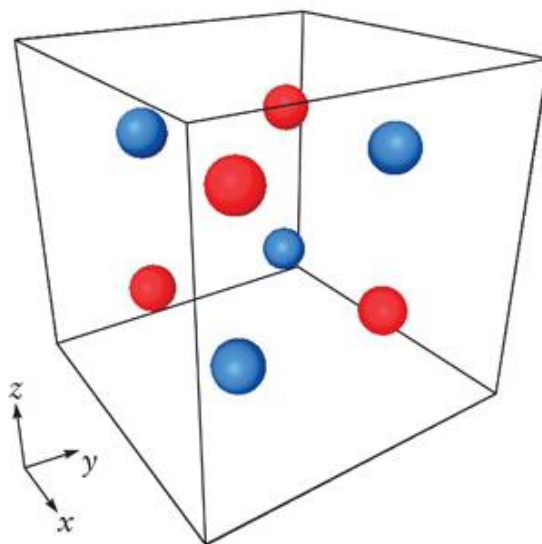
(Image: <https://commons.wikimedia.org/wiki/File:Particle-in-a-box-3d.svg>)

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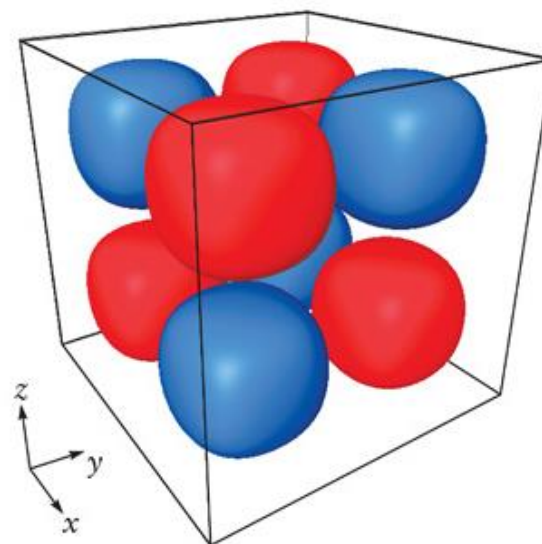
Contour Surfaces



$|\Psi| = 0.9$

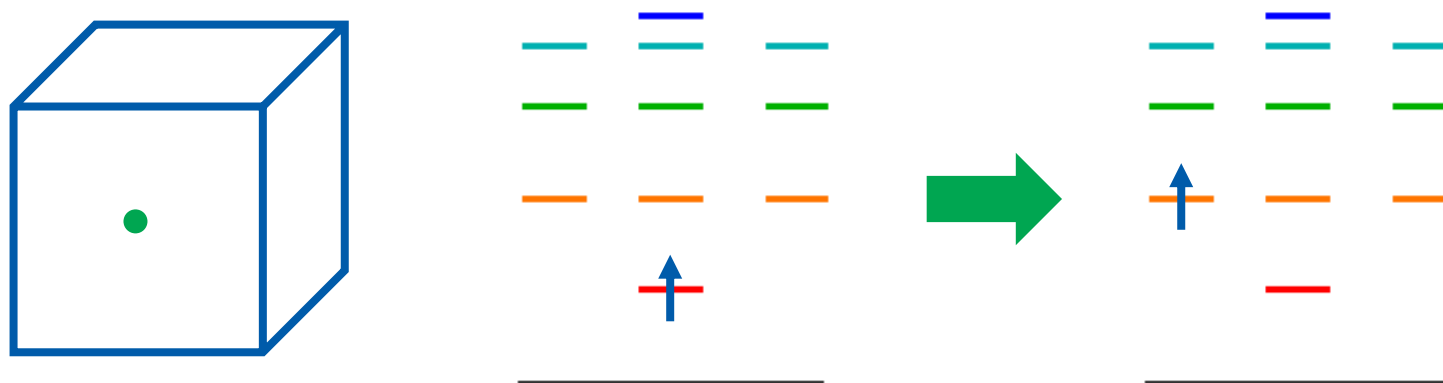


$|\Psi| = 0.3$



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Application: Molecular Translation



$$\Delta E = \frac{h^2}{8ma^2} (6 - 3) = \frac{3h^2}{8ma^2} = \frac{hc}{\lambda}$$

For a He atom (6.646×10^{-27} kg) in a box with $a = 0.1$ m,

$$\Delta E = 2.48 \times 10^{-39} \text{ J} = 1.49 \times 10^{-18} \text{ kJ/mol}$$

$$\lambda = 8.02 \times 10^{13} \text{ m: undetectable!}$$